

One world, two ways to add a source

A, B and U are independent fair bits: eight worlds, each with probability $1/8$.

Held fixed in both comparisons

Existing source: A

Target: $Y = (A,B)$

Candidate U: independent extra bit

Sources (A,U); target (A,B)

Eight observations; each has mass $1/8$

The source pair and target together
identify all three original bits.

MGW synergy: $\log(4/3)$

Added information after A: 0

Candidate B: complete the target

Sources (A,B); target (A,B)

Four observations; each has mass $2/8$

For each fixed (A,B), worlds with
 $U=0$ and $U=1$ merge into one output.

MGW synergy: $\log(4/3)$

Added information after A: $\log 2$

Added information here is conditional mutual information after A. All values are in nats.